

CAPSTONE PROJECT

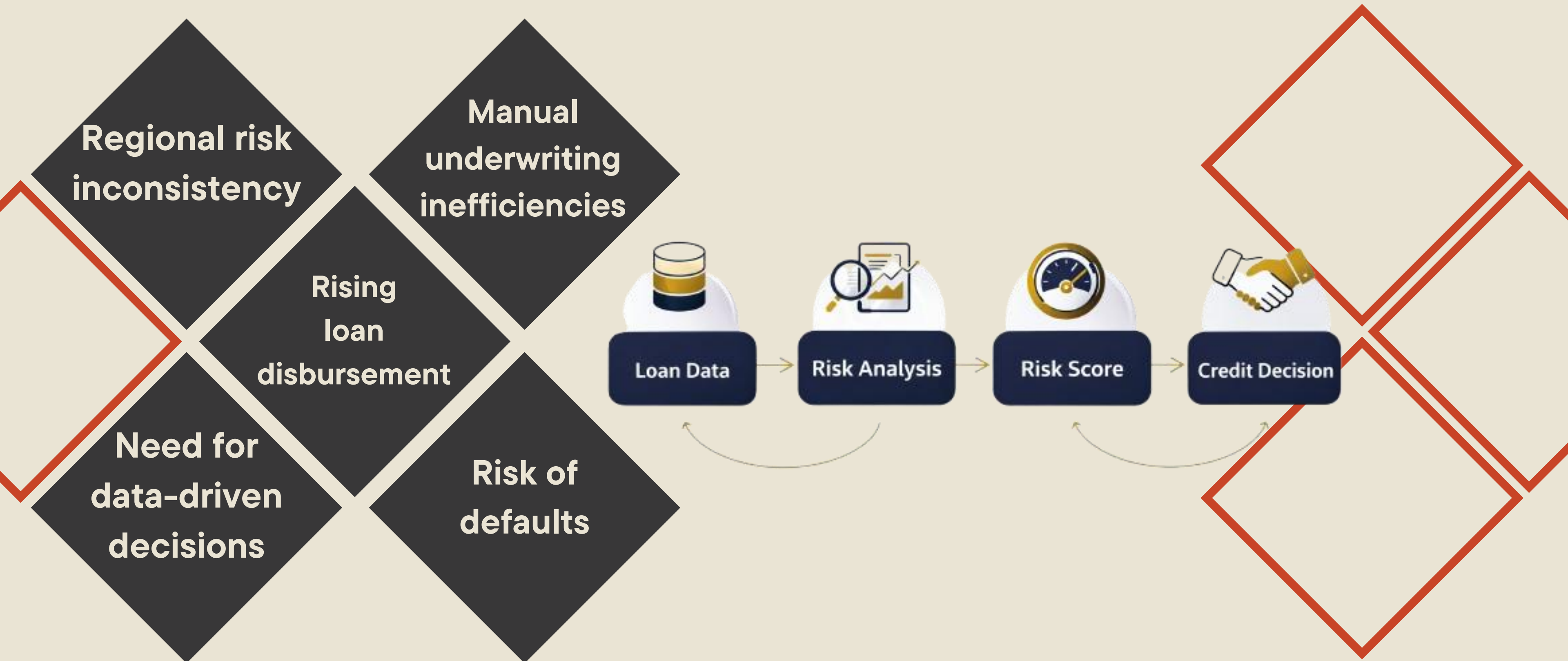
GOLDMAN SACHS CONSUMER FINANCE ANALYSIS

SQL Analysis • Power BI Dashboard • Machine Learning

By: Vaishnavi Chakraborty



BUSINESS CHALLENGE





DATASET OVERVIEW

Customer	270,299 records
	Loan
Region merged	28 initial features
	Approval rate: 91%
Highly imbalanced dataset	Default rate: 0.044%

CAPSTONE PROJECT



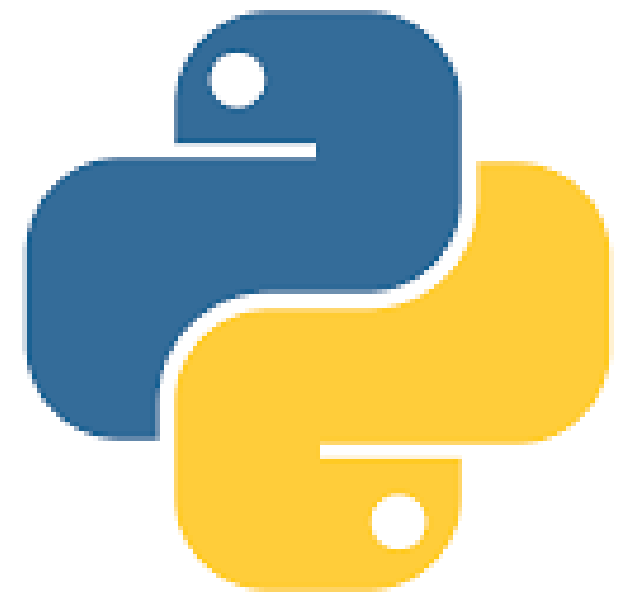
**ADVANCED
SQL ANALYSIS**

PART 1



**INTERACTIVE
DASHBOARDS**

PART 2



**MACHINE
LEARNING**

PART 3

ADVANCED SQL ANALYSIS

PART 1





UNIQUE CUSTOMERS BY REGION & AVG CREDIT SCORE



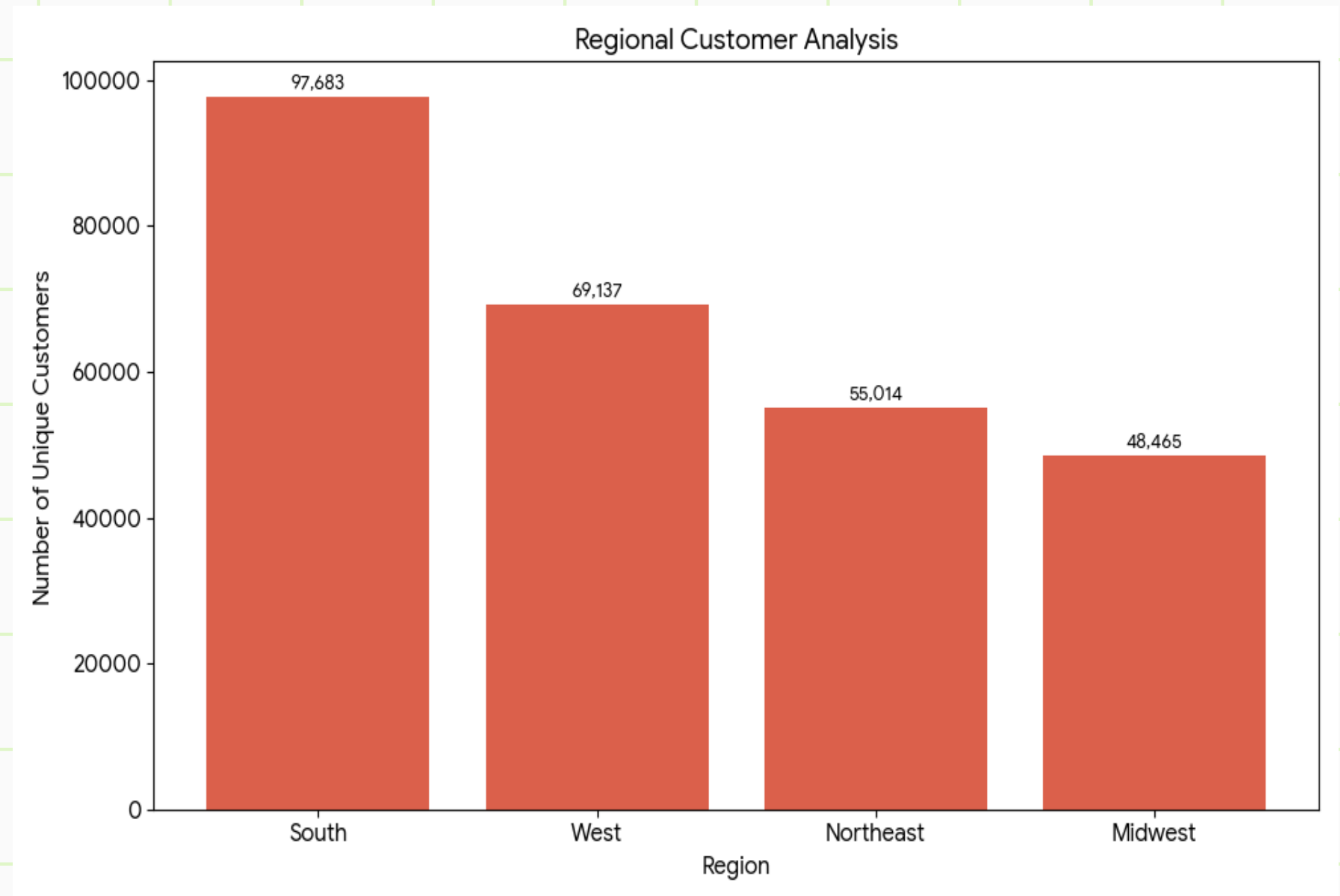
REGION	CUSTOMERS	AVG_CRED_SCR
South	97,683	2.6087
West	69,137	2.5675
Northeast	55,014	2.5742
Midwest	48,465	2.5731

Midwest - 48,465
Avg Score: 2.57

South - 97,683
Avg Score: 2.60

Northeast - 55,014
Avg Score: 2.57

West - 69,137
Avg Score: 2.56



South region leads in customer concentration, reflecting stronger lending demand



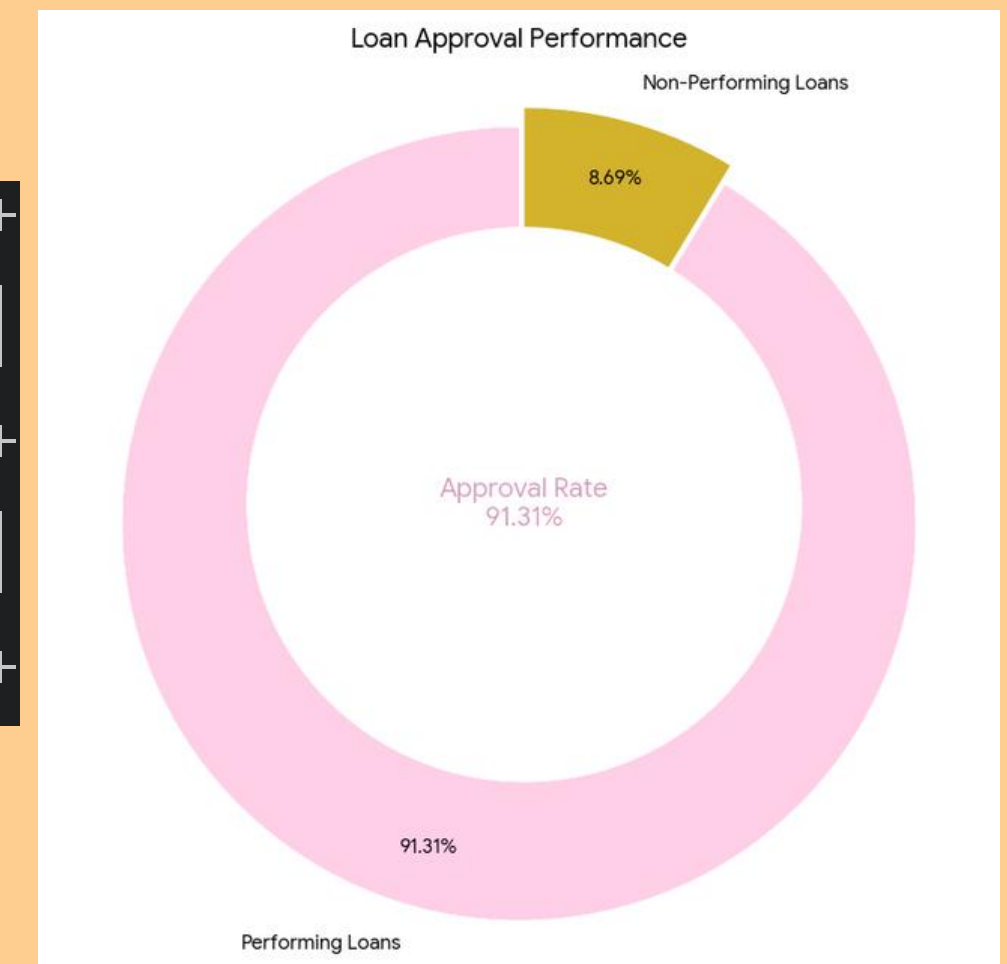
LOAN APPROVAL EFFICIENCY ✨

```
SELECT
  COUNT(*) AS total_loans,
  SUM(CASE
    WHEN loan_status IN ('Fully Paid', 'Current')
    THEN 1 ELSE 0
  END) AS performing_loans,
  ROUND(
    SUM(CASE
      WHEN loan_status IN ('Fully Paid', 'Current')
      THEN 1 ELSE 0
    END) * 100.0 / COUNT(*),
    2
  ) AS performing_rate_percentage
FROM loans;
```

Total Loans: 270,299

```
+-----+
| TOTAL_LOANS |
+-----+
| 270,299    |
+-----+
+-----+
| PERF_RATE%  |
+-----+
| 91.31       |
+-----+
```

```
+-----+
| PERFORMING  |
+-----+
| 246,822    |
+-----+
```



Performing Loans: 246,822

The portfolio demonstrates strong repayment performance with limited default exposure

HIGH DEFAULT RISK IDENTIFICATION



High Risk Loan Criteria:

A loan is flagged as High Risk when:

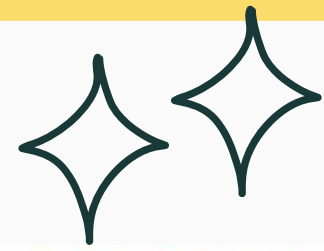
- Annual Income is below portfolio average
- Funded Amount is above portfolio average
- Credit Grade falls under D, E, F, or G

```
WITH risk_base AS (  
  SELECT  
    l.loan_id,  
    l.customer_id,  
    c.annual_inc,  
    l.grade,  
    l.funded_amount,  
    AVG(c.annual_inc) OVER () AS avg_income,  
    AVG(l.funded_amount) OVER () AS avg_funded  
  FROM loan l  
  JOIN customers c  
    ON l.customer_id = c.customer_id  
)
```

```
SELECT *,  
  CASE  
    WHEN annual_inc < avg_income  
      AND funded_amount > avg_funded  
      AND grade IN ('D','E','F','G')  
    THEN 'High Default Risk'  
    ELSE 'Low / Moderate Risk'  
  END AS risk_flag
```

+-----+-----+-----+		
TOTAL_LOANS	PERFORMING	PERF_RATE%
+-----+-----+-----+		
270,299	246,822	91.31
+-----+-----+-----+		

Weak credit quality combined with high loan exposure significantly increases default probability.

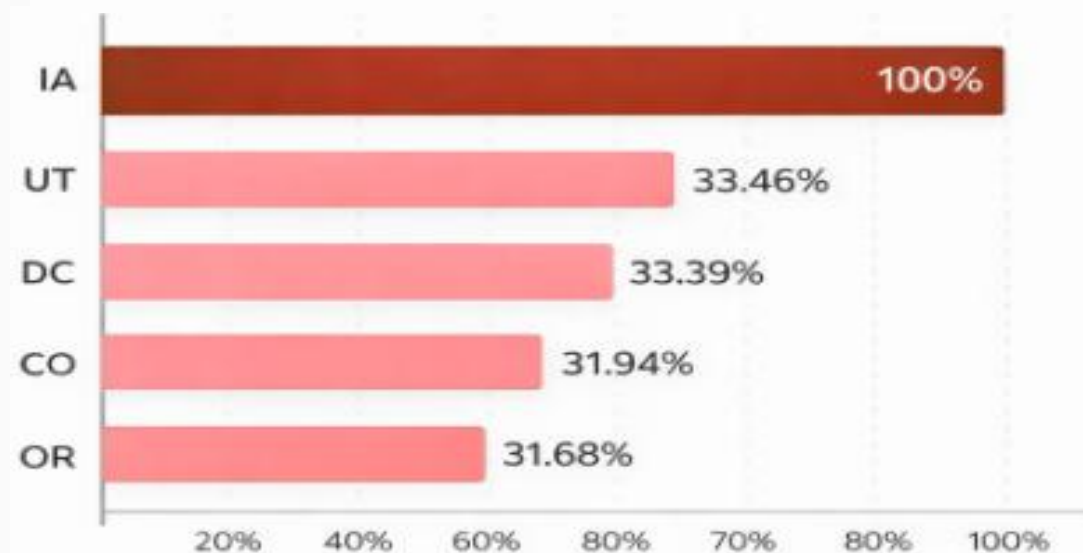


TOP 5 STATES BY APPROVAL RATE



```
WITH loan_summary AS (SELECT state,COUNT(*) AS total_loans,
SUM(CASE WHEN loan_status = 'Fully Paid' THEN 1 ELSE 0 END) AS approved_loans from loan
GROUP BY state),
avg_approved_amount AS (
SELECT state,ROUND(AVG(loan_amount), 2) AS avg_approved_loan_amount FROM loan
WHERE loan_status = 'Fully Paid' GROUP BY state )
SELECT ls.state, ls.total_loans,ls.approved_loans,
ROUND((ls.approved_loans * 100.0 / ls.total_loans), 2) AS approval_rate_percentage,
aa.avg_approved_loan_amount FROM loan_summary ls
JOIN avg_approved_amount aa
ON ls.state = aa.state
ORDER BY approval_rate_percentage DESC
LIMIT 5;
```

TOP 5 STATES BY APPROVAL RATE

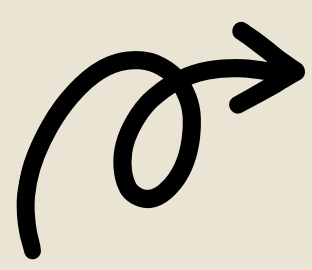


Average Approved Loan Amount (Top States)

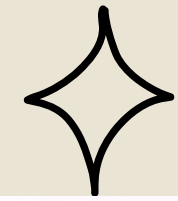
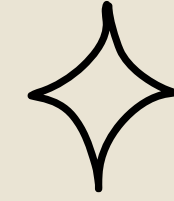
IA	\$6,725
UT	\$14,598
DC	\$14,146
CO	\$14,600
OR	\$13,139

state	total_loans	approved_loans	approval_rate_percentage	avg_approved_loan_amount
IA	1	1	100.00	6725.00
UT	1781	596	33.46	14598.36
DC	626	209	33.39	14145.69
CO	5732	1831	31.94	14600.11
OR	3324	1053	31.68	13138.56

State-level variations indicate regional policy or borrower profile differences



YEARLY LOAN GROWTH TRENDS - CAR LOANS



```
WITH yearly_purpose AS (SELECT issue_year,purpose,COUNT(*) AS total_loans FROM loan
GROUP BY issue_year, purpose )
SELECT issue_year,purpose,total_loans,
LAG(total_loans) OVER (PARTITION BY purpose ORDER BY issue_year) AS prev_year_loans,
ROUND((total_loans - LAG(total_loans) OVER (PARTITION BY purpose ORDER BY issue_year))
/ LAG(total_loans) OVER (PARTITION BY purpose ORDER BY issue_year) * 100,2) AS growth_percentage
FROM yearly_purpose
ORDER BY purpose, issue_year;
```

issue_year	purpose	total_loans	prev_year_loans	growth_percentage
2012	car	41	NULL	NULL
2013	car	106	41	158.54
2014	car	190	106	79.25
2015	car	346	190	82.11
2016	car	467	346	34.97

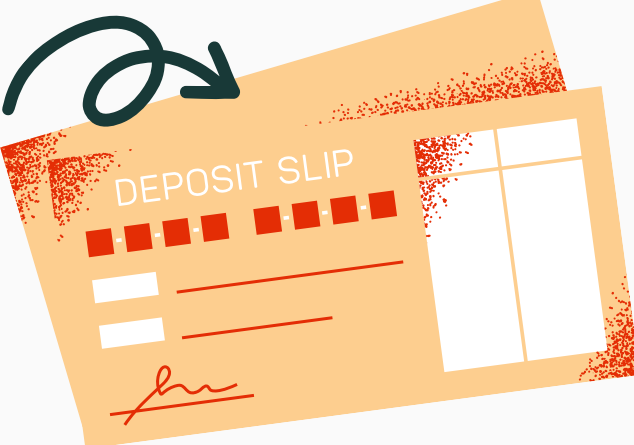
YEARLY CAR LOAN GROWTH TRENDS



Car Loans Growth Example:

- 2013: +158%
- 2014: +79%
- 2015: +82%
- 2016: +35%

Car loans show strong year-over-year expansion, signaling growing consumer financing demand.



AVERAGE LOANS PER CUSTOMER & REPAYMENT HISTORY



AVERAGE LOANS PER CUSTOMER: 1.00

REPAYMENT CATEGORIES:

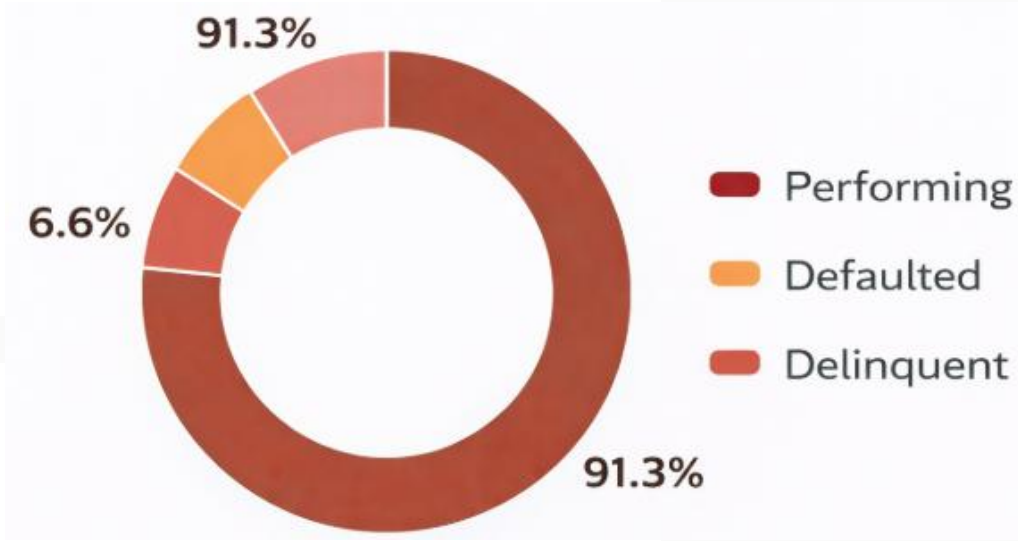
PERFORMING - 91.31%

DEFAULTED - 6.61%

DELINQUENT - 2.08%

```
WITH loans_per_customer AS (SELECT customer_id,COUNT(*) AS total_loans FROM loan
GROUP BY customer_id),
avg_loans AS (SELECT ROUND(AVG(total_loans), 2) AS avg_loans_per_customer FROM loans_per_customer),
repayment_summary AS (SELECT
CASE
WHEN loan_status IN ('Fully Paid', 'Current')
THEN 'Performing'
WHEN loan_status IN ('Late (16-30 days)', 'Late (31-120 days)', 'In Grace Period')
THEN 'Delinquent'
WHEN loan_status IN ('Default', 'Charged Off')
THEN 'Defaulted'
END AS repayment_category
FROM loan
)

SELECT
r.repayment_category,
COUNT(*) AS total_loans,
ROUND(COUNT(*) * 100.0 / (SELECT COUNT(*) FROM loan), 2) AS percentage_share,
a.avg_loans_per_customer
FROM repayment_summary r
CROSS JOIN avg_loans a
GROUP BY r.repayment_category, a.avg_loans_per_customer;
```



repayment_category	total_loans
Performing	246822
Defaulted	17863
Delinquent	5614

percentage_share	avg_loans_per_customer
91.31	1.00
6.61	1.00
2.08	1.00

Most customers take a single loan, indicating low repeat borrowing behavior within the dataset.

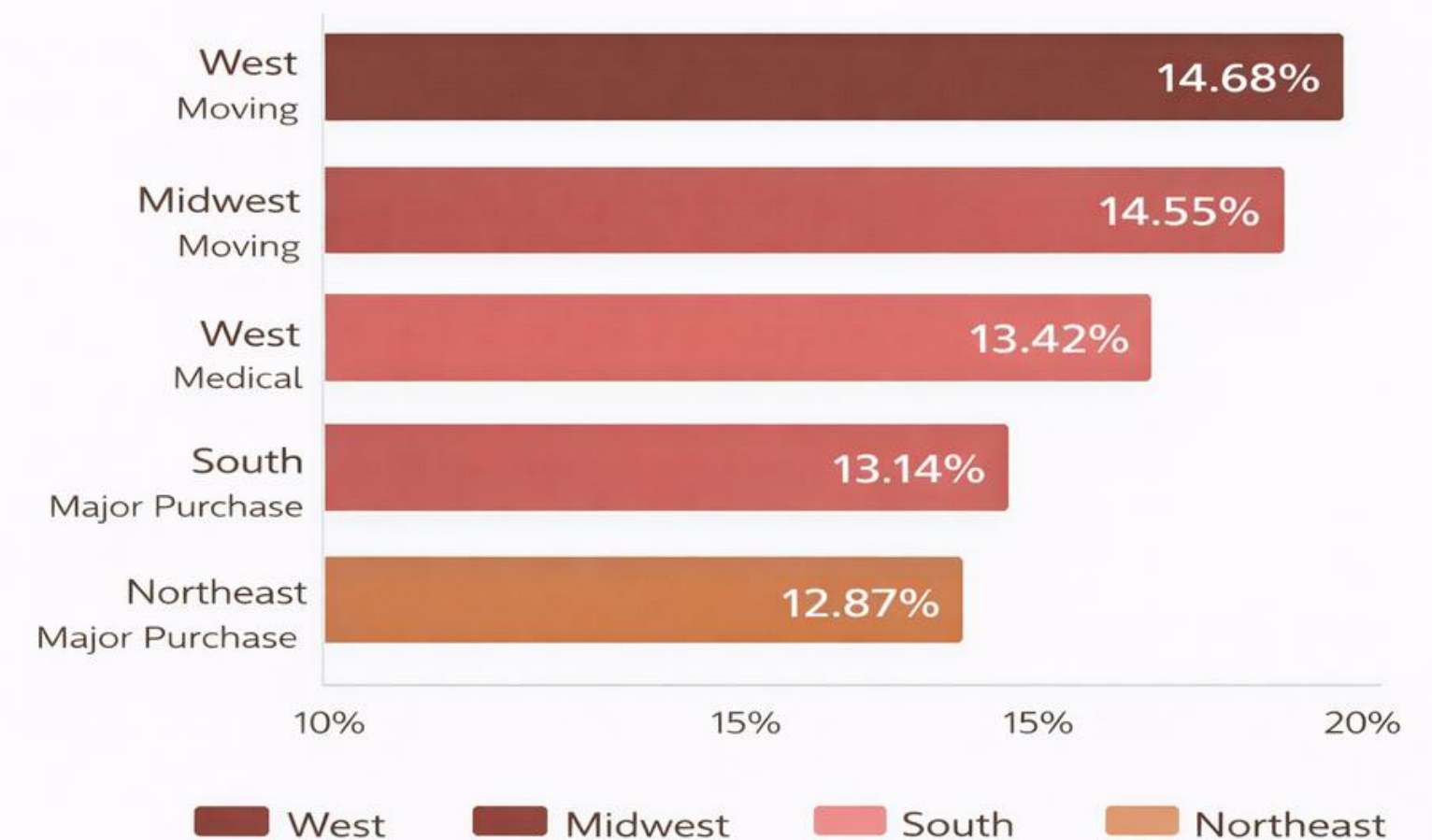
AVERAGE INTEREST RATE BY CATEGORY & REGION

```
select * from loan;

select s.region,l.purpose,round(avg(l.int_rate) * 100, 2) as avg_int_rate_percentage
from loan l join state_region s
on l.state=s.state
group by l.purpose,s.region;
```

region	purpose	avg_int_rate_percentage
Midwest	moving	14.55
West	moving	14.68
West	medical	13.42
Northeast	major_purchase	12.87
South	major_purchase	13.14

Average Interest Rate by Region & Loan Purpose



HIGHER RATES IN MOVING-RELATED LOANS MAY REFLECT HIGHER SHORT-TERM RISK EXPOSURE COMPARED TO STRUCTURED MAJOR PURCHASE LOANS

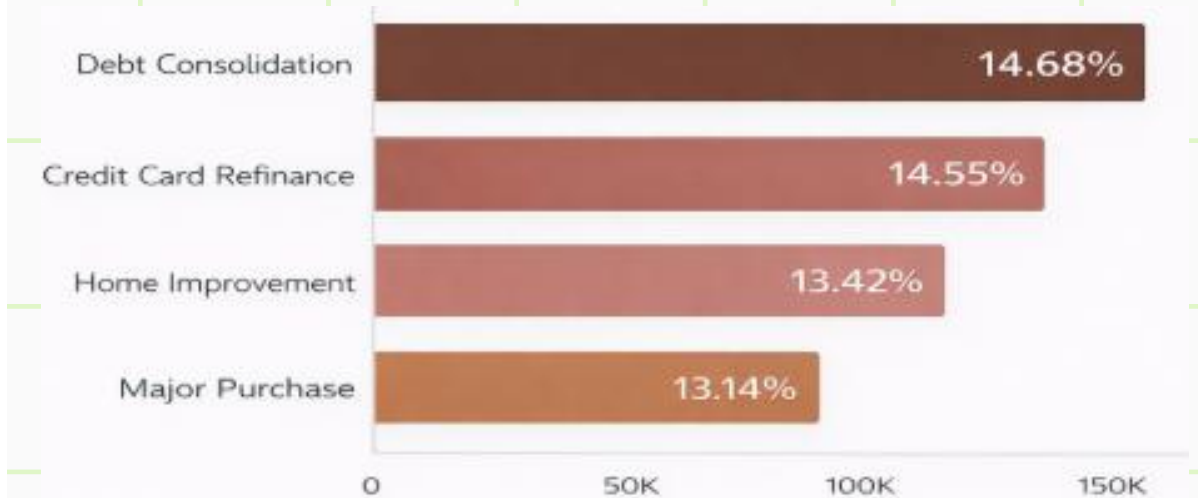


LOAN PURPOSE POPULARITY & DEFAULT RATE

```
select purpose, sum(case when loan_status='Default' then 1 else 0 End)/count(*)*100 as Default_Rate
from loan
group by purpose
order by count(*) desc
limit 1;
```

```
SELECT
    purpose,
    SUM(CASE WHEN loan_status = 'Default' THEN 1 ELSE 0 END)
    / COUNT(*) * 100 AS default_rate
FROM loan
GROUP BY purpose
ORDER BY COUNT(*) DESC
LIMIT 1;
```

purpose	default_rate
debt_consolidation	0.0039



MOST COMMON LOAN PURPOSE:

DEBT CONSOLIDATION

DEFAULT RATE: 0.39%



INCOME LEVEL VS APPROVAL RATE



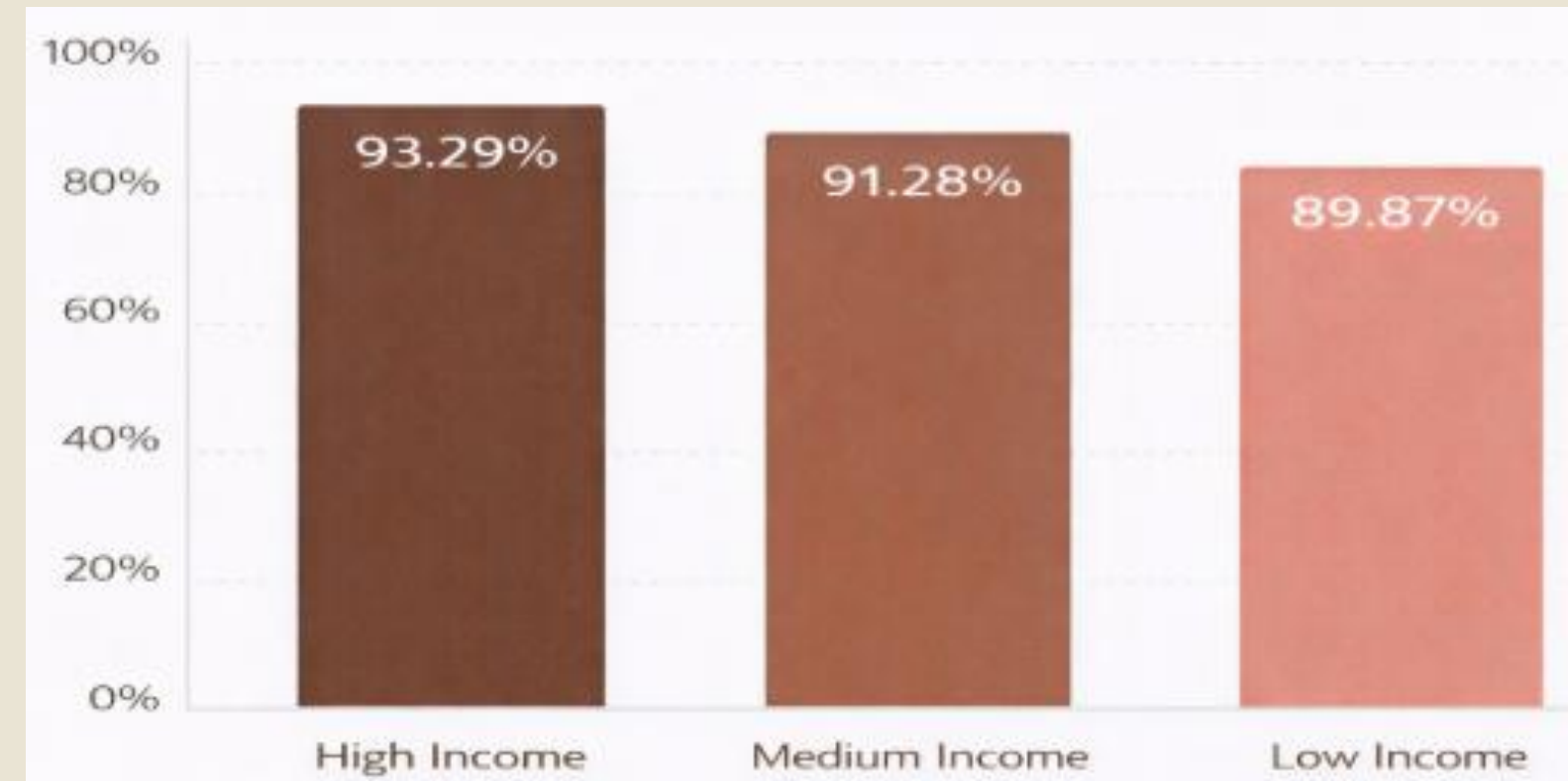
```
select * from customers;
select * from loan;

with income_category as (select c.customer_id,
| case when annual_inc<50000 then 'Low'
  when annual_inc BETWEEN 50000 and 100000 then 'Medium'
  Else 'High'
  End as 'Income_Level',l.loan_status
  from customers c join loan l
  on c.customer_id=l.customer_id)
Select Income_Level, round(sum(case when loan_status IN ('Fully Paid','Current') then 1 else 0
  end )/count(*)*100,2) as approval_rate_percentage

from income_category
group by Income_Level
order by approval_rate_percentage desc;
```

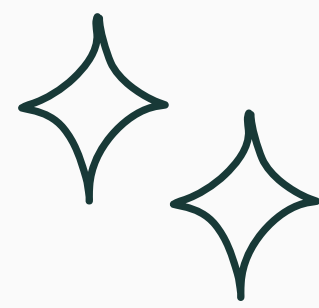
Income_Level	approval_rate_percentage
High	93.29
Medium	91.28
Low	89.87

Income Level vs Approval Rate

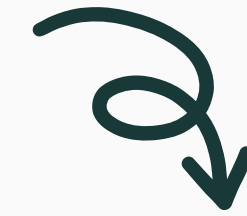


93.29% APPROVAL RATE
HIGH INCOME SEGMENT

Approval likelihood increases with income level, validating income as a key credit determinant



CUSTOMERS WITH MISSED PAYMENTS EARLY RISK DETECTION



```
select distinct loan_status from loan;
with risky_loans as (select customer_id, loan_status from loan
                     where loan_status not in ('Fully Paid', 'Current'))
select customer_id, count(*) as missed_payments from risky_loans
group by customer_id
having count(*) >= 1
order by missed_payments DESC;
```

customer_id	missed_payments
b' \xd3i\xd6\xdb8\x80t*g\xa...	1
b' \xda\xe0\x9e\xd6\xbf\xe...	1
b' \x8eV\xb7\xa6u~}\x84\x...	1
b' ()K\x11\xff\x06aQY0\x1f?...	1
b' \xe6(\x98\x03\xb3!\xf7B...	1

Risk Alert Flow Diagram



Identifying borrowers showing early stress signals before full default

Customers were flagged when:

Loan status NOT IN ('Fully Paid', 'Current')

Grouped by Customer ID

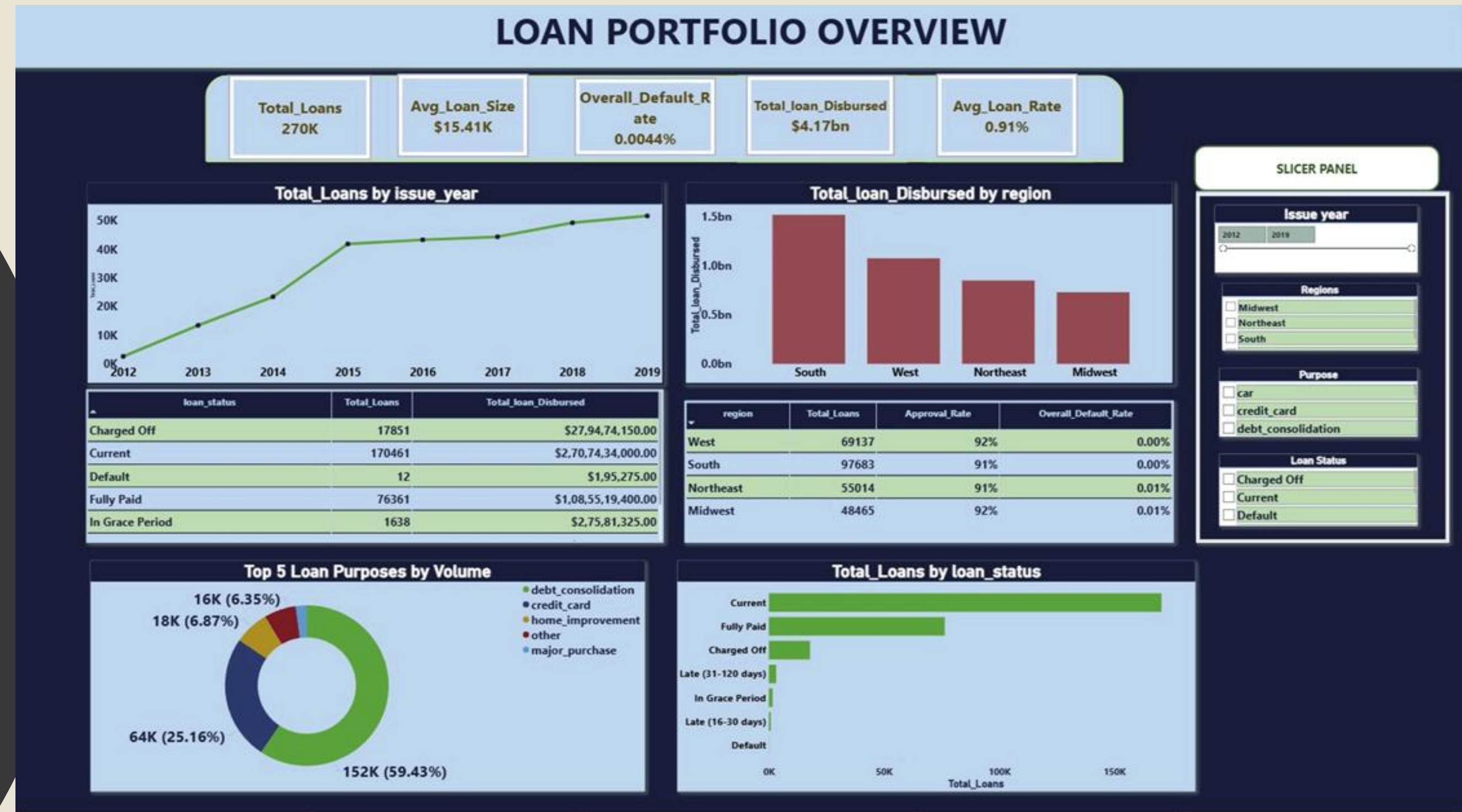
Missed payments ≥ 1

INTERACTIVE RISK MONITORING DASHBOARDS

PART 2



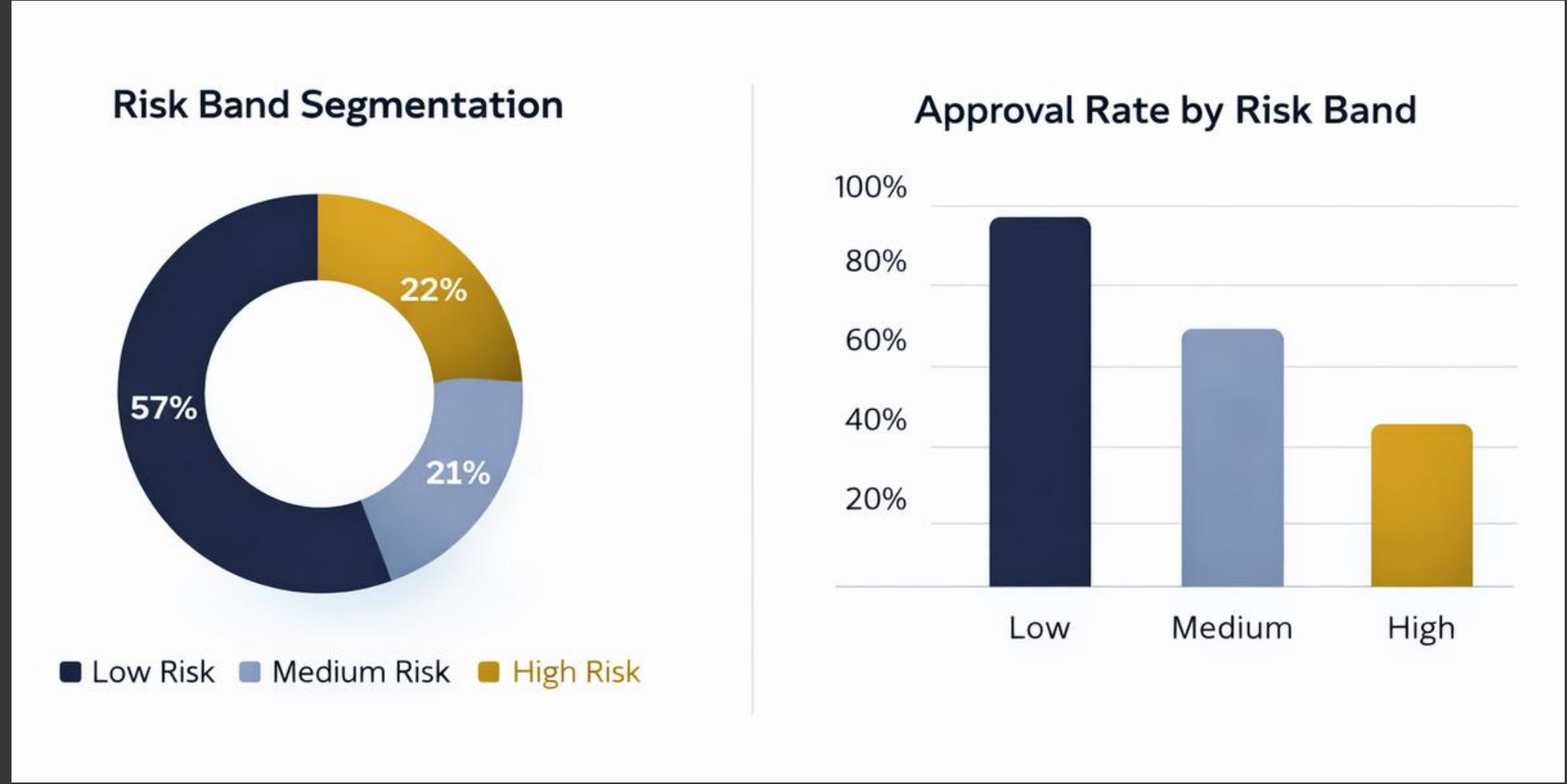
LOAN PORTFOLIO OVERVIEW



- Real-time KPI monitoring
- Regional risk segmentation
- Loan category performance insights

TOTAL LOANS: 270K
LOAN DISBURSED: \$4.17BN
DEFAULT RATE: 0.0044%

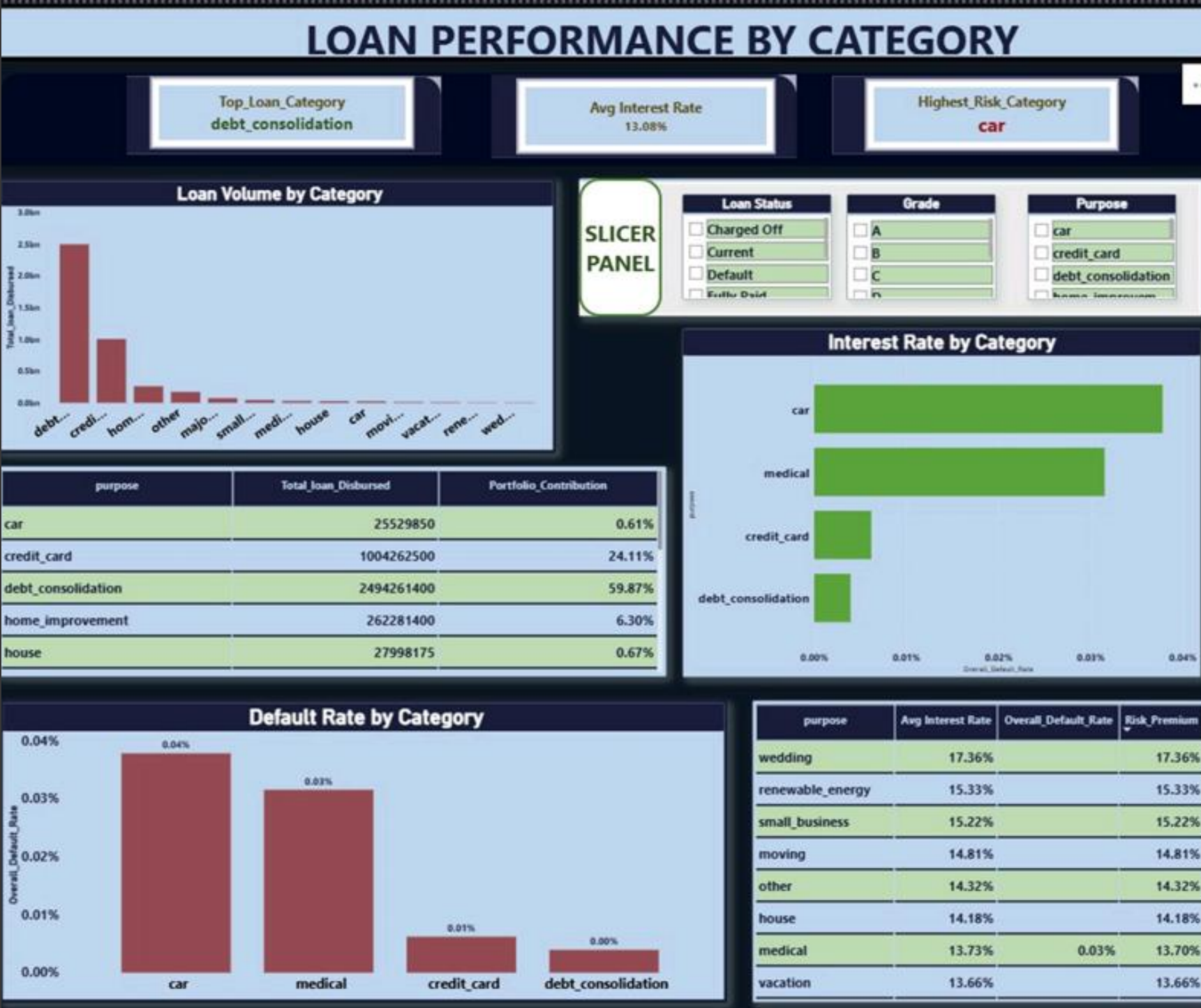
CUSTOMER RISK SEGMENTATION & APPROVAL PATTERN



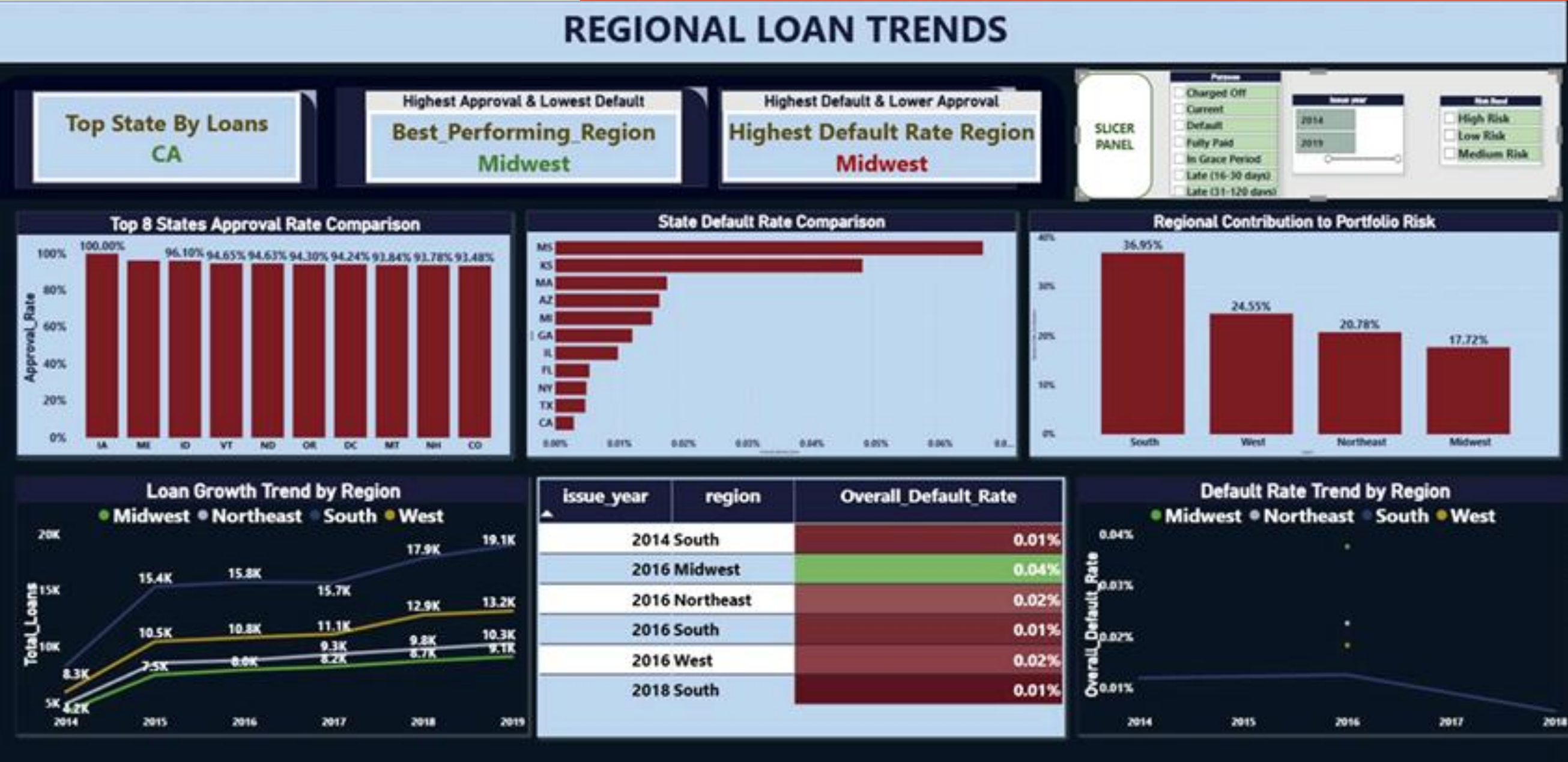
HIGH RISK GROUP SHOWS HIGHER REJECTION PATTERN.

LOAN CATEGORY PERFORMANCE & RISK EXPOSURE

WHILE DEBT CONSOLIDATION
DOMINATES PORTFOLIO VOLUME, CAR
AND MEDICAL LOANS EXHIBIT
COMPARATIVELY HIGHER DEFAULT
RISK.



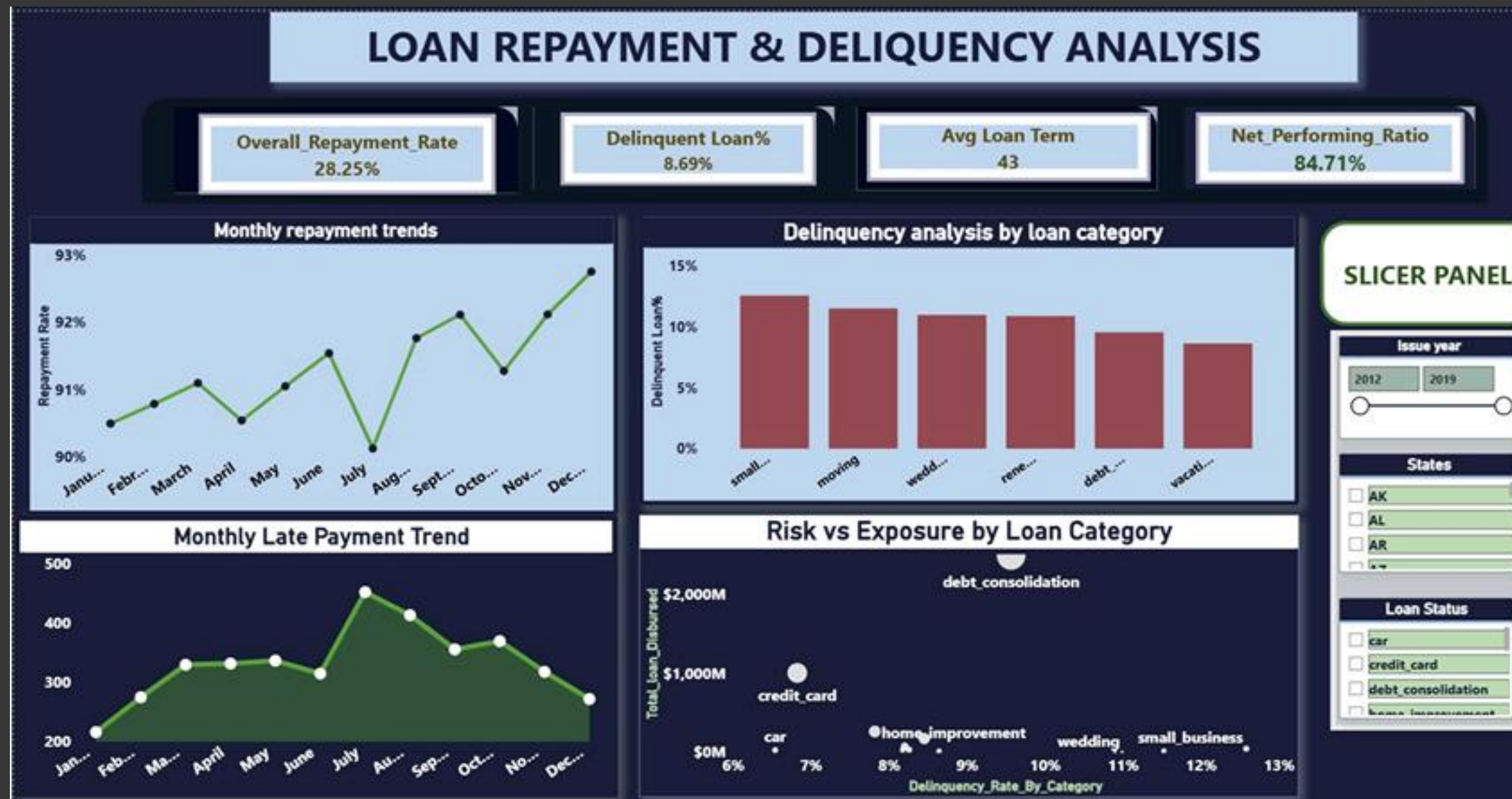
REGIONAL RISK EXPOSURE & DEFAULT TRENDS



The **South** region exhibits the **highest** overall portfolio **risk contribution**, driven by both loan volume and elevated default exposure.



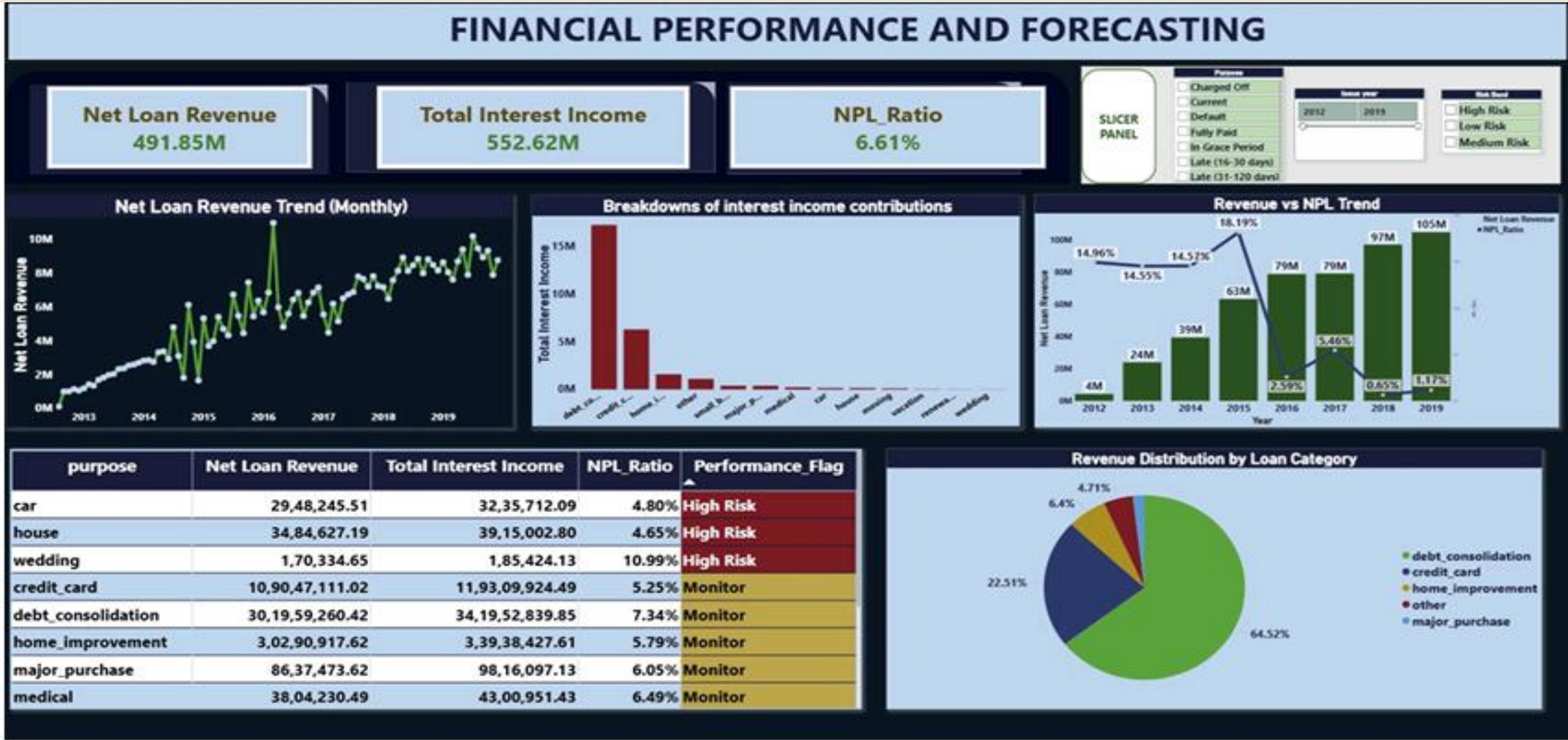
LOAN CATEGORY PERFORMANCE & RISK EXPOSURE ANALYSIS



Delinquency levels **exhibit seasonal spikes during mid-year, indicating potential cyclical repayment stress**

REPAYMENT RATE = 28.25%
DELINQUENT LOANS = 8.69%
NET PERFORMING RATIO = 84.71%

FINANCIAL PERFORMANCE & RISK STABILITY



LOAN REVENUE DEMONSTRATES STEADY GROWTH, WHILE NPL RATIO EXHIBITS PERIODIC FLUCTUATIONS, HIGHLIGHTING THE NEED FOR PROACTIVE RISK MANAGEMENT.

MACHINE LEARNING

MODEL 1 AND MODEL 2

PART 3



LIBRARIES IMPLEMENTED

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder, StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import classification_report, confusion_matrix
```

Built two classification models:

Loan Approval Prediction and Loan Default Prediction

DATA COLLECTION & INTEGRATION

Data Sources

Loan Dataset

Customer Dataset

Regional Dataset

▼ Data Exploration

Loaded customer, loan, and region datasets to begin structured exploratory analysis.

```
[7]: cust=pd.read_csv(r"C:\Users\Shreya\OneDrive\Desktop\WsCubeTech\Data Analytics Mentorship Program\CAPSTONE-PROJECT\Goldman Sachs\customer.csv")
      loan=pd.read_csv(r"C:\Users\Shreya\OneDrive\Desktop\WsCubeTech\Data Analytics Mentorship Program\CAPSTONE-PROJECT\Goldman Sachs\loan.csv")
      loan_count_year=pd.read_csv(r"C:\Users\Shreya\OneDrive\Desktop\WsCubeTech\Data Analytics Mentorship Program\CAPSTONE-PROJECT\Goldman Sachs\loan_count_year.csv")
      loan_purpose=pd.read_csv(r"C:\Users\Shreya\OneDrive\Desktop\WsCubeTech\Data Analytics Mentorship Program\CAPSTONE-PROJECT\Goldman Sachs\loan_purpose.csv")
      loan_with_region=pd.read_csv(r"C:\Users\Shreya\OneDrive\Desktop\WsCubeTech\Data Analytics Mentorship Program\CAPSTONE-PROJECT\Goldman Sachs\loan_with_region.csv")
      state_region=pd.read_csv(r"C:\Users\Shreya\OneDrive\Desktop\WsCubeTech\Data Analytics Mentorship Program\CAPSTONE-PROJECT\Goldman Sachs\state_region.csv")
```

Used .info(),describe(),shape() to understand data types, structure, distributions, and detect missing values.



LOAN APPROVAL PREDICTION

OVERVIEW

```
# Create modeling dataset
df_model = df.copy()
```

```
# Drop columns not needed for ML
```

```
df_model = df_model.drop(columns=[
    "state",
    "type"
])
```

```
# Confirm structure
df_model.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 270299 entries, 0 to 270298
Data columns (total 16 columns):
 #   Column              Non-Null Count  Dtype  
---  -
 0   loan_amount         270299 non-null float64
 1   funded_amount       270299 non-null float64
 2   term                270299 non-null int64  
 3   int_rate            270299 non-null float64
 4   installment         270299 non-null float64
 5   grade               270299 non-null object
 6   issue_year         270299 non-null float64
 7   pymnt_plan          270299 non-null bool  
 8   purpose             270299 non-null object
 9   emp_length          270299 non-null float64
10  home_ownership       270299 non-null object
11  annual_inc          270299 non-null float64
12  verification_status  270299 non-null object
13  avg_cur_bal         270299 non-null float64
14  Tot_cur_bal         270299 non-null float64
15  approval_status     270299 non-null int64  
dtypes: bool(1), float64(9), int64(2), object(4)
memory usage: 31.2+ MB
```

```
df.select_dtypes(include="object").columns
```

```
Index(['state', 'grade', 'type', 'purpose', 'home_ownership',
      'verification_status'],
      dtype='object')
```

```
# One-hot encode categorical features
X = pd.get_dummies(X, drop_first=True)

print("New X shape:", X.shape)
X.head()
```

New X shape: (270299, 36)

	loan_amount	funded_amount	term	int_rate	installment	issue_year	pymnt_plan	emp_length	annual_inc	avg_cur_bal	...	purpose_small_business	pu
0	1000.0	1000.0	36	0.2295	38.689999	2013.0	False	0.0	84000.0	3209.0	...	False	
1	1200.0	1200.0	36	0.1629	42.369999	2014.0	False	0.0	32600.0	3068.0	...	False	
2	1400.0	1400.0	36	0.1114	45.930000	2012.0	False	5.0	37000.0	6351.0	...	False	
3	1975.0	1975.0	36	0.1049	64.190002	2014.0	False	8.0	29000.0	2667.0	...	False	
4	2000.0	2000.0	36	0.1875	73.059998	2013.0	False	3.0	60000.0	3580.0	...	False	

5 rows x 36 columns



LOAN APPROVAL

```
log_model_balanced = LogisticRegression(max_iter=1000, class_weight="balanced")  
  
log_model_balanced.fit(X_train_scaled, y_train)  
  
y_pred_bal = log_model_balanced.predict(X_test_scaled)
```

```
print("Confusion Matrix:\n", confusion_matrix(y_test, y_pred_bal))  
print("\nClassification Report:\n", classification_report(y_test, y_pred_bal))
```

Confusion Matrix:

```
[[ 3301  1394]  
 [15679 33686]]
```

Classification Report:

	precision	recall	f1-score	support
0	0.17	0.70	0.28	4695
1	0.96	0.68	0.80	49365
accuracy			0.68	54060
macro avg	0.57	0.69	0.54	54060
weighted avg	0.89	0.68	0.75	54060

Balanced Logistic Regression

Model successfully identifies **70%** of risky loan applications.

- Accuracy: 68%
- Risk Recall: 70%
- Improved minority detection

Predicted Safe
34,000

Predicted Risk
8,000

3,000

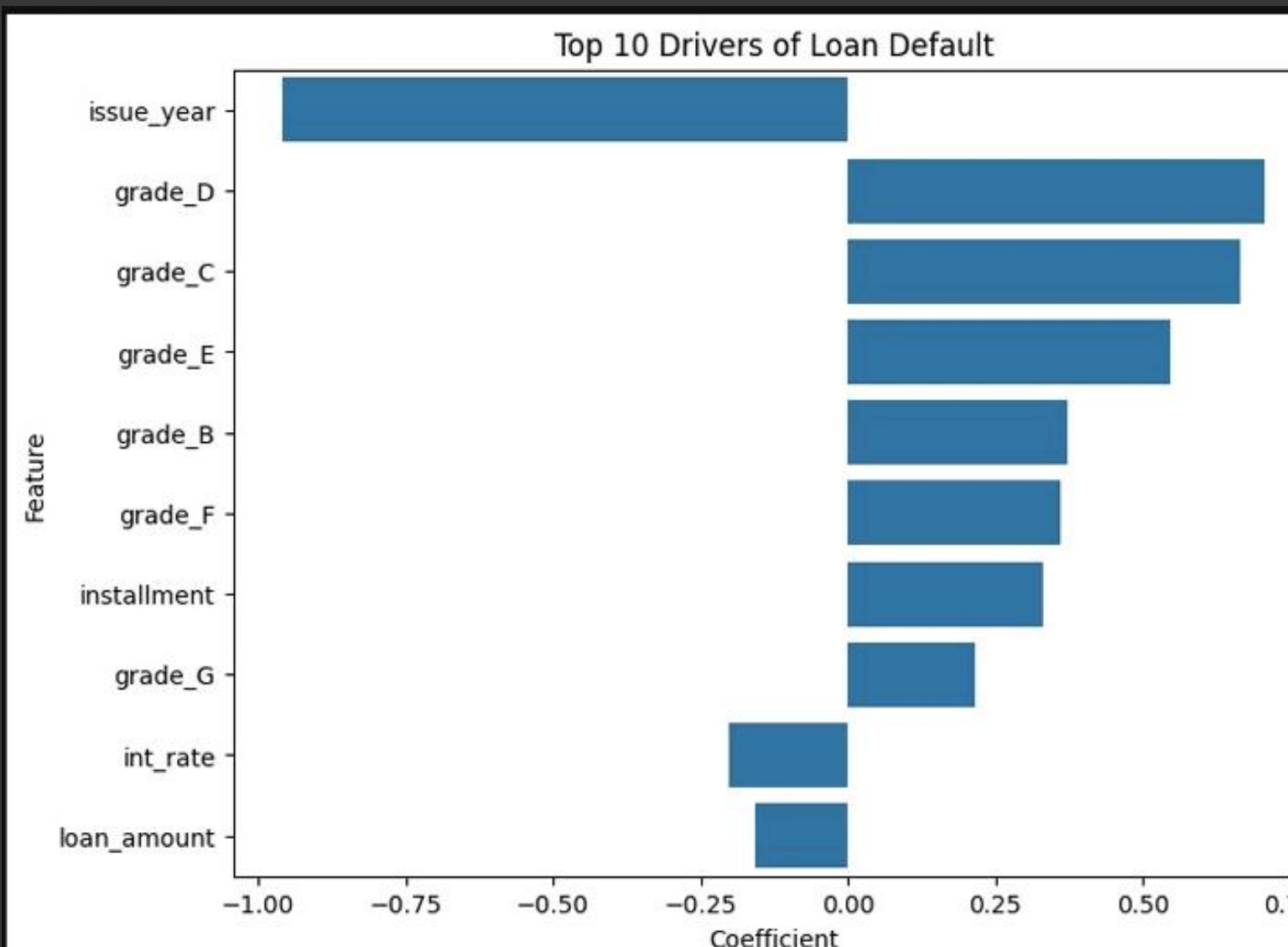
7,000

APPROVAL DRIVERS



Loan Grade

Top Drivers of Loan Approval



```
# Get top 10 absolute impact features
coef_def_df["Abs_Coefficient"] = coef_def_df["Coefficient"].abs()
top_features = coef_def_df.sort_values(by="Abs_Coefficient", ascending=False).head(10)

plt.figure(figsize=(8,6))
sns.barplot(
    x="Coefficient",
    y="Feature",
    data=top_features
)

plt.title("Top 10 Drivers of Loan Default")
plt.show()
```

Top Influencing Factors:

Loan Grade – **strongest determinant**

Installment burden affects approval likelihood

Employment stability increases approval probability

Mortgage ownership positively influences decision

DEFAULT PREDICTION

Model 2

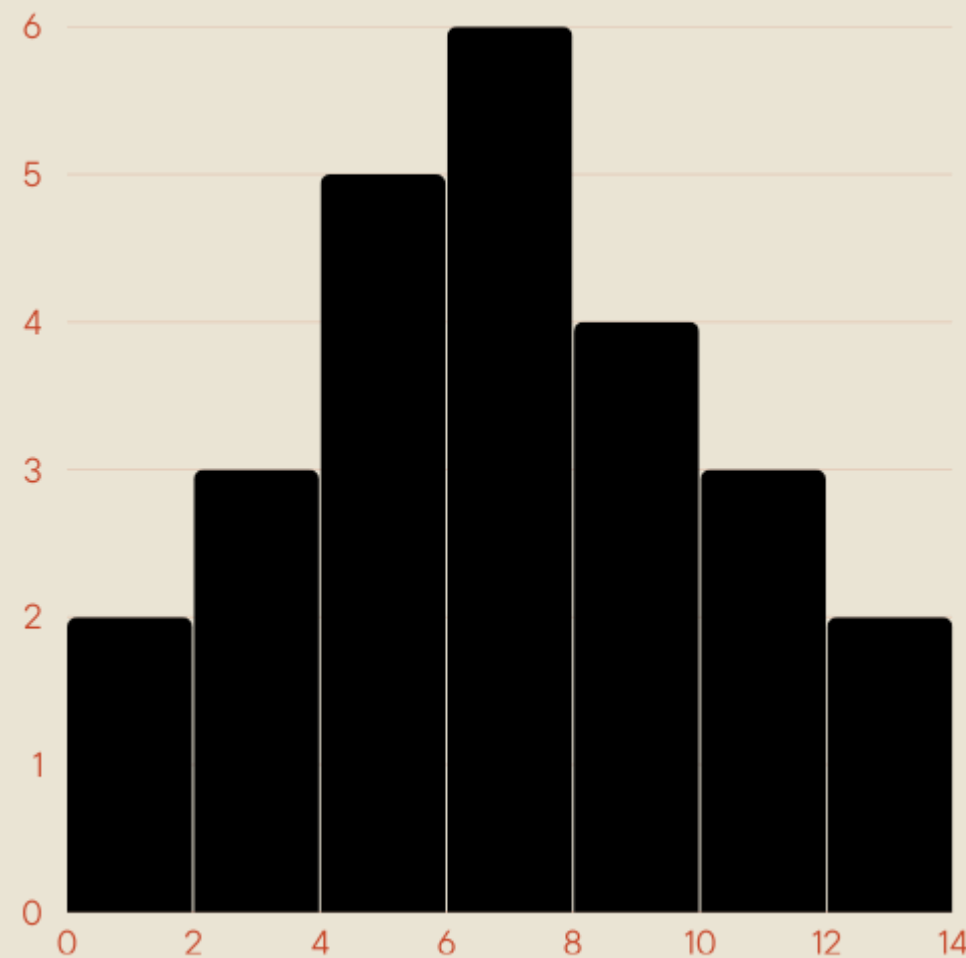
Addressing Class Imbalance

Class Distribution

- Non-Default $\approx$ 252K
- Default $\approx$ 17K
- Default Rate $\approx$ 6.6%

Key Challenge

- Highly imbalanced dataset
- Recall prioritized over accuracy



Loan Default Prediction Model

```
# Recreate clean merged dataset
df_base = loan.merge(cust, on="customer_id", how="left")

print(df_base.shape)
```

```
(270299, 28)
```

```
df_default = df_base.copy()

df_default["default_status"] = df_default["loan_status"].apply(
    lambda x: 1 if x in ["Default", "Charged Off"] else 0
)

df_default["default_status"].value_counts()
```

```
default_status
0    252436
1     17863
Name: count, dtype: int64
```

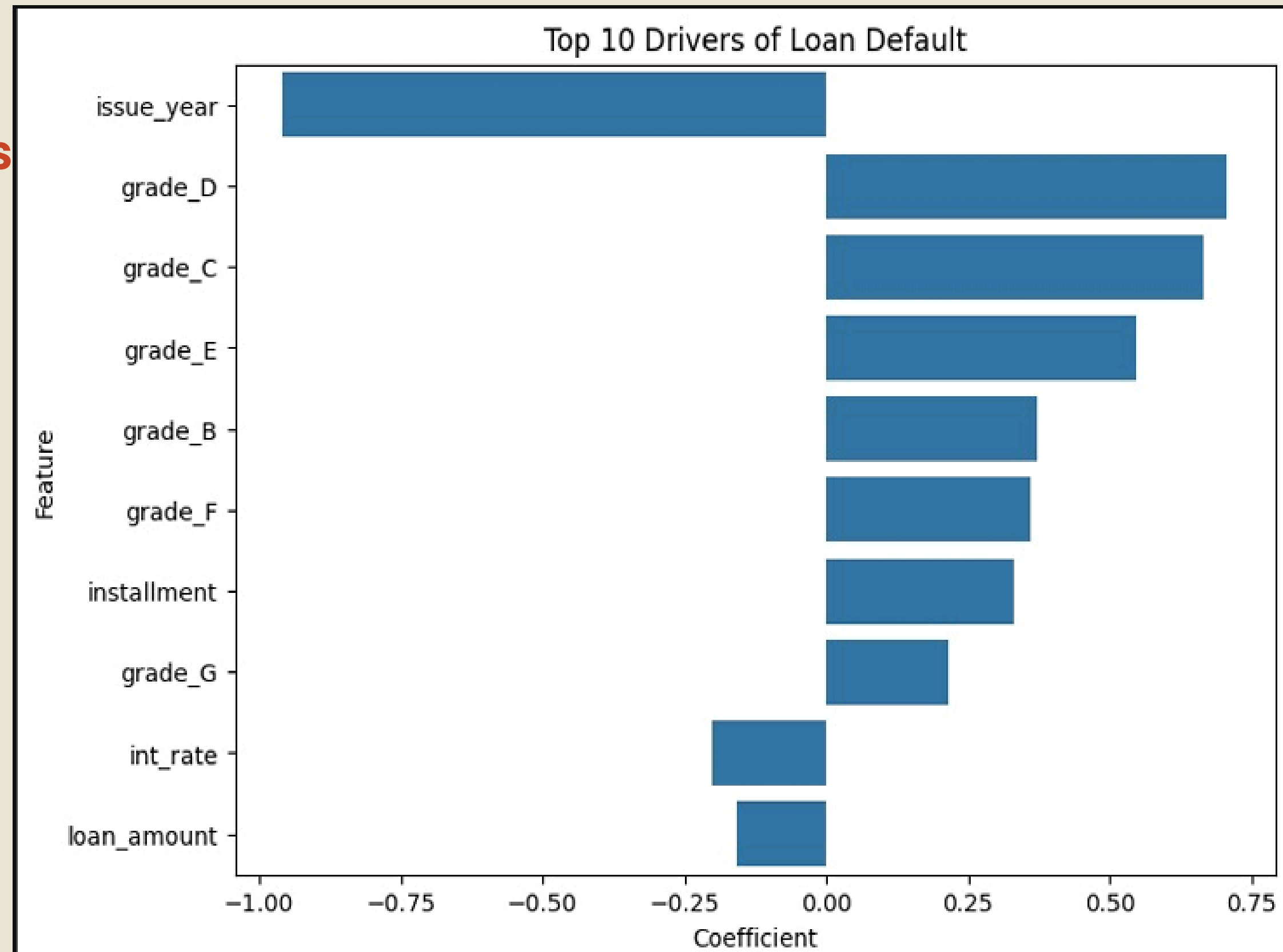

DEFAULT RISK DRIVERS

Model Interpretability – Risk Factors

Visually emphasize:

- **Loan Grade**
- **Installment**
- **Debt Consolidation**

Lower loan grades and higher installment burdens significantly increase default probability.



MODEL COMPARISON

Approval vs Default Prediction

Confusion Matrix:
[[3301 1394]
[15679 33686]]

Classification Report:		precision	recall	f1-score	support
	0	0.17	0.70	0.28	4695
	1	0.96	0.68	0.80	49365
accuracy				0.68	54060
macro avg		0.57	0.69	0.54	54060
weighted avg		0.89	0.68	0.75	54060

Confusion Matrix:
[[35870 14617]
[827 2746]]

Classification Report:		precision	recall	f1-score	support
	0	0.98	0.71	0.82	50487
	1	0.16	0.77	0.26	3573
accuracy				0.71	54060
macro avg		0.57	0.74	0.54	54060
weighted avg		0.92	0.71	0.79	54060

Metric	Approval Model	Default Model	Both models prioritize recall to effectively identify high-risk borrowers.
Recall	70%	77%	
Accuracy	68%	71%	

BUSINESS IMPACT OF PREDICTIVE MODELING

Executive Summary – Loan Risk Modeling

This project developed two supervised machine learning models to support Global FinBank's risk management strategy:

1 Loan Approval Prediction

Balanced Logistic Regression achieved 70% recall for identifying risky loan applications.

Loan grade emerged as the strongest determinant of approval.

Employment stability, mortgage ownership, and higher financial balances positively influenced approval probability.

2 Loan Default Prediction

Balanced Logistic Regression achieved 77% recall in detecting actual defaults.

Loan grade, installment burden, and debt consolidation purpose significantly increased default risk.

Employment length, higher account balances, and recent loan issuance reduced default probability.

Business Impact

The models demonstrate the ability to proactively flag high-risk borrowers while maintaining structured interpretability. These insights can help loan officers refine approval criteria and strengthen risk mitigation strategies.

FROM DATA TO PREDICTIVE INTELLIGENCE

This project integrates SQL-based data analysis, interactive Power BI dashboards, and machine learning models to transform credit risk management into a proactive intelligence system. By combining descriptive insights with predictive modeling, the solution enhances underwriting accuracy, reduces default exposure, and enables data-driven credit decisions that strengthen long-term portfolio stability.

FROM REACTIVE MONITORING TO PROACTIVE RISK INTELLIGENCE.



THANK YOU

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